

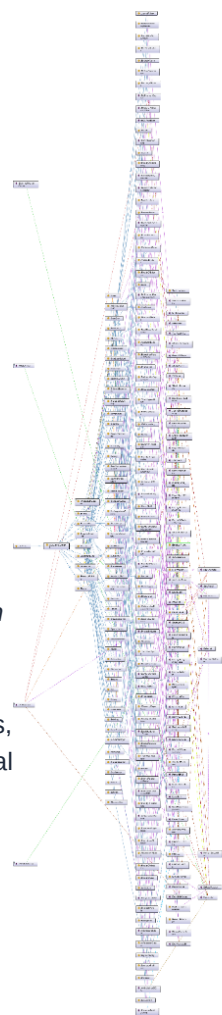
Academic Organization

Semantic Web OWL Ontology Structure & Instance Report

Author:	Murtuza Yusuf Rangwala (Matricola: VR50566)
Professor:	Matteo Cristani
Ontology URI:	http://www.example.org/academicorg
Source File:	C:/Users/murta/Desktop/AcademicOrganization.owl
Total RDF Triples:	716
Classes:	145
Object Properties:	15
Datatype Properties:	8
Named Individuals:	81

Executive Summary

This report outlines the structural model and loaded instances of the *Academic Organization* ontology. The ontology provides a semantic framework representing university structures, including person taxonomies (students, academic and administrative staff, alumni), organizational units (faculties, departments, research centers), academic programs, courses, events, assessments, grants, and facilities. It enables complex relational querying and logical inference for academic domain management.



1. Class Hierarchy (Taxonomy)

Below is the complete taxonomic inheritance tree of the ontology classes. The hierarchy starts from the root class **AcademicConcept**. Subclasses inherit properties defined on their parent classes.

```
AcademicConcept
├── AcademicEvent
│   ├── Colloquium
│   ├── Conference
│   │   ├── InternationalConference
│   │   └── NationalConference
│   ├── DefenseExamination
│   ├── Graduation
│   ├── Hackathon
│   ├── OpenDay
│   └── Symposium
├── AcademicProgram
│   ├── CertificateProgram
│   ├── DegreeProgram
│   │   ├── BachelorProgram
│   │   ├── DoctoralProgram
│   │   └── MastersProgram
│   ├── DiplomaProgram
│   ├── ExchangeProgram
│   └── OnlineProgram
├── Assessment
│   ├── LabAssessment
│   ├── OralExam
│   ├── PeerReviewAssessment
│   ├── ProjectAssessment
│   ├── QuizAssessment
│   └── WrittenExam
├── Course
│   ├── CoreCourse
│   ├── ElectiveCourse
│   ├── GraduateCourse
│   ├── Lab
│   ├── OnlineCourse
│   ├── Seminar
│   ├── Thesis
│   ├── UndergraduateCourse
│   └── Workshop
└── Facility
    └── Building
```

- AcademicBuilding
 - ResidenceHall
 - Cafeteria
 - Classroom
 - Auditorium
 - LectureHall
 - ComputerLaboratory
 - ScientificLaboratory
 - DryLaboratory
 - WetLaboratory
 - SportsFacility
- FinancialAid
 - Bursary
 - ErasmusGrant
 - ResearchFellowship
 - Scholarship
 - MeritScholarship
 - NeedBasedScholarship
 - TeachingFellowship
- OrganizationalUnit
 - Department
 - BiologyDepartment
 - ChemistryDepartment
 - ComputerScienceDepartment
 - EconomicsDepartment
 - LinguisticsDepartment
 - MathematicsDepartment
 - PhysicsDepartment
 - Faculty
 - FacultyOfEconomics
 - FacultyOfEngineering
 - FacultyOfHumanities
 - FacultyOfLaw
 - FacultyOfMedicine
 - FacultyOfScience
 - GraduateSchool
 - Institute
 - LanguageInstitute
 - Library
 - ResearchCenter
 - AIResearchCenter
 - BiomedicalResearchCenter
 - ClimateResearchCenter
 - StudentUnion
 - University

```

    ■■■ PrivateUniversity
    ■■■ PublicUniversity
    ■■■ ResearchUniversity
    ■■■ TechnicalUniversity
■■■ Person
    ■■■ AcademicStaff
        ■■■ Adjunct
        ■■■ Lecturer
            ■■■ SeniorLecturer
        ■■■ Professor
            ■■■ AssistantProfessor
            ■■■ AssociateProfessor
            ■■■ EmeritusProfessor
            ■■■ FullProfessor
        ■■■ Researcher
            ■■■ PostdoctoralResearcher
            ■■■ PrincipalInvestigator
            ■■■ ResearchAssociate
        ■■■ TeachingAssistant
        ■■■ VisitingScholar
    ■■■ AdministrativeStaff
        ■■■ AdmissionsOfficer
        ■■■ Dean
        ■■■ DepartmentHead
        ■■■ FinanceOfficer
        ■■■ ITStaff
        ■■■ LibraryStaff
        ■■■ Rector
        ■■■ Secretary
        ■■■ ViceRector
    ■■■ Alumni
        ■■■ BachelorAlumni
        ■■■ MastersAlumni
        ■■■ PhDAlumni
    ■■■ Student
        ■■■ DistanceLearningStudent
        ■■■ ExchangeStudent
        ■■■ FullTimeStudent
        ■■■ GraduateStudent
            ■■■ MastersStudent
            ■■■ PhDStudent
        ■■■ InternationalStudent
        ■■■ PartTimeStudent
        ■■■ UndergraduateStudent
■■■ ResearchActivity

```

- ■ ■ Grant
 - ■ ■ GovernmentGrant
 - ■ ■ EUGrant
 - ■ ■ PrivateGrant
- ■ ■ Patent
- ■ ■ Publication
 - ■ ■ BookChapter
 - ■ ■ ConferencePaper
 - ■ ■ JournalArticle
 - ■ ■ Monograph
 - ■ ■ Preprint
 - ■ ■ TechnicalReport
- ■ ■ ResearchProject
 - ■ ■ CollaborativeResearchProject
 - ■ ■ FundedResearchProject
 - ■ ■ InternalResearchProject

2. Ontology Properties

2.1 Object Properties

Object properties establish semantic relationships between instances of two classes. The table below details each object property, along with its domain (starting class) and range (target class).

Property Name	Domain Class	Range Class
attendsEvent	Person	AcademicEvent
authors	AcademicStaff	Publication
belongsTo	Person	OrganizationalUnit
conductsResearch	Researcher	ResearchProject
enrolledIn	Student	Course
fundedBy	ResearchProject	Grant
heldIn	Course	Facility
manages	AdministrativeStaff	OrganizationalUnit
partOf	Department	Faculty
participatesIn	Person	Assessment
produces	ResearchProject	Publication
receives	Student	FinancialAid
registeredIn	Student	AcademicProgram
supervisedBy	PhDStudent	Professor
teaches	AcademicStaff	Course

2.2 Datatype Properties

Datatype properties map individuals to literal values (numbers, strings, floats, etc.). The table below defines the properties, their domains, and literal types.

Property Name	Domain Class	Literal Range Type
hasAmount	FinancialAid	float
hasCapacity	Facility	integer

hasCreditHours	Course	integer
hasEmail	Person	string
hasGrade	Assessment	float
hasName	AcademicConcept	string
hasStudentID	Student	string
hasYearEstablished	OrganizationalUnit	integer

3. Named Individuals (Instances)

The ontology contains concrete instances representing real-world academic objects. Below is a structured list of these individuals, categorized by their primary classes, including their property assertions and connections.

Institutions & Organizational Units

Individual Name	Class Type	Properties & Relationships
AILabVerona	AIResearchCenter	hasName: Artificial Intelligence Lab, University of Verona
CSAILMit	AIResearchCenter	hasName: Computer Science and Artificial Intelligence Laboratory, MIT
DeptBiologyOxford	BiologyDepartment	hasName: Department of Biology, University of Oxford
BiomedCenterVerona	BiomedicalResearchCenter	hasName: Biomedical Research Center, University of Verona
DeptCSVerona	ComputerScienceDepartment	hasName: Department of Computer Science, University of Verona; partOf: FacultyOfScienceVerona
DeptEconomicsStanford	EconomicsDepartment	hasName: Department of Economics, Stanford University
FacultyOfEngineeringPolimi	FacultyOfEngineering	hasName: Faculty of Engineering, Politecnico di Milano
FacultyOfHumanitiesOxford	FacultyOfHumanities	hasName: Faculty of Humanities, University of Oxford
FacultyOfMedicineVerona	FacultyOfMedicine	hasName: Faculty of Medicine, University of Verona
FacultyOfScienceVerona	FacultyOfScience	hasName: Faculty of Science, University of Verona
DeptMathVerona	MathematicsDepartment	hasName: Department of Mathematics, University of Verona; partOf: FacultyOfScienceVerona
DeptPhysicsMIT	PhysicsDepartment	hasName: Department of Physics, MIT
StanfordUniversity	PrivateUniversity	hasName: Stanford University; hasYearEstablished: 1891
OxfordUniversity	PublicUniversity	hasName: University of Oxford; hasYearEstablished: 1096
UniversityOfVerona	PublicUniversity	hasName: University of Verona; hasYearEstablished: 1982
MITUniversity	ResearchUniversity	hasName: Massachusetts Institute of Technology; hasYearEstablished: 1861
PolitecnicoMilano	TechnicalUniversity	hasName: Politecnico di Milano; hasYearEstablished: 1863

Academic Personnel & Students

Individual Name	Class Type	Properties & Relationships
DrFerrari	AssistantProfessor	belongsTo: DeptCSVerona; hasEmail: laura.ferrari@univr.it; hasName: Dr. Laura Ferrari
ProfGambara	AssociateProfessor	belongsTo: DeptMathVerona; hasEmail: matteo.gambara@univr.it; hasName: Prof. Matteo Gambara
DeanMarini	Dean	hasEmail: dean.science@univr.it; hasName: Prof. Anna Marini; manages: FacultyOfScienceVerona
EmmaGreen	DistanceLearningStudent	hasEmail: emma.green@online.ox.ac.uk; hasName: Emma Green; hasStudentID: OX20230042
ProfEmeritusConti	EmeritusProfessor	hasEmail: roberto.conti@univr.it; hasName: Prof. Roberto Conti
LucaMartini	ExchangeStudent	hasEmail: luca.martini@erasmus.eu; hasName: Luca Martini; hasStudentID: ERAS001; receives: ErasmusMobilityGrant; registeredIn: ErasmusProgramVerona
ProfCastellani	FullProfessor	attendsEvent: CVPRConference2025, AISymposiumVerona; belongsTo: DeptCSVerona; hasEmail: umberto.castellani@univr.it; hasName: Prof. Umberto Castellani
ProfHawkins	FullProfessor	belongsTo: DeptPhysicsMIT; hasEmail: a.hawkins@mit.edu; hasName: Prof. Alan Hawkins
ITSupportTech	ITStaff	hasEmail: it.support@univr.it; hasName: Carlo Ferretti
AishaKhan	InternationalStudent	belongsTo: DeptCSVerona; hasEmail: aisha.khan@studenti.univr.it; hasName: Aisha Khan; hasStudentID: VR654321; receives: MeritScholarshipAI
LibrarianFranco	LibraryStaff	hasEmail: biblioteca@univr.it; hasName: Franco Neri
AlumniMariaDonati	MastersAlumni	hasEmail: maria.donati@alumni.univr.it; hasName: Maria Donati

MurtuzaRangwala	MastersStudent	attendsEvent: AIIHackathon2025; belongsTo: DeptCSVerona; enrolledIn: HCI_Course, MLFinance_Course; hasEmail: murtuza.rangwala@studenti.univr.it; hasName: Murtuza Rangwala; hasStudentID: VR50566; participatesIn: HCI_FinalExam, MLFinance_ProjectAssessment; registeredIn: MSAIVerona; supervisedBy: ProfCastellani
AlumniJamesWong	PhDAumni	hasEmail: j.wong@alumni.mit.edu; hasName: James Wong
SaraRossi	PhDStudent	belongsTo: DeptMathVerona; enrolledIn: LinearAlgebra_Course; hasEmail: sara.rossi@phd.univr.it; hasName: Sara Rossi; hasStudentID: VR789012; registeredIn: PhDMathVerona; supervisedBy: ProfGambara
DrPatelPostdoc	PostdoctoralResearcher	belongsTo: AILabVerona; hasEmail: priya.patel@univr.it; hasName: Dr. Priya Patel
RectorBianchi	Rector	hasEmail: rettore@univr.it; hasName: Prof. Pier Bianchi; manages: UniversityOfVerona
SecretaryGiulia	Secretary	belongsTo: DeptCSVerona; hasEmail: segreteria.informatica@univr.it; hasName: Giulia Esposito
TAMarco	TeachingAssistant	belongsTo: DeptCSVerona; hasEmail: marco.bianchi@ta.univr.it; hasName: Marco Bianchi
JohnSmith	UndergraduateStudent	belongsTo: DeptPhysicsMIT; enrolledIn: QuantumPhysics_Course; hasEmail: j.smith@mit.edu; hasName: John Smith; hasStudentID: MIT998877; participatesIn: PhysicsMidterm
VisitorDrNakamura	VisitingScholar	belongsTo: AILabVerona; hasEmail: k.nakamura@visiting.univr.it; hasName: Dr. Kenji Nakamura

Courses & Academic Programs

Individual Name	Class Type	Properties & Relationships
BSComputerScienceMIT	BachelorProgram	hasName: Bachelor of Science in Computer Science, MIT
OnlineDataScienceCert	CertificateProgram	hasName: Certificate in Data Science, Stanford Online
DataStructures_Course	CoreCourse	hasCreditHours: 9; hasName: Data Structures and Algorithms
LinearAlgebra_Course	CoreCourse	hasCreditHours: 9; hasName: Linear Algebra; teaches: ProfGambara
PhDMathVerona	DoctoralProgram	hasName: PhD Programme in Mathematics, University of Verona
ComputerVision_Course	ElectiveCourse	hasCreditHours: 6; hasName: Computer Vision and 3D Reconstruction; teaches: ProfCastellani
ErasmusProgramVerona	ExchangeProgram	hasName: Erasmus+ Exchange Programme, University of Verona
HCI_Course	GraduateCourse	hasCreditHours: 6; hasName: HCI - Intelligent Multimodal Interfaces; heldIn: LectureHallAlpha; teaches: ProfCastellani
MLFinance_Course	GraduateCourse	hasCreditHours: 6; hasName: Applied Machine Learning in Finance; teaches: ProfGambara
MSAIVerona	MastersProgram	hasName: Master's Degree in Artificial Intelligence, University of Verona
ResearchMethodsSeminar	Seminar	hasCreditHours: 3; hasName: Research Methods in Computer Science
QuantumPhysics_Course	UndergraduateCourse	hasCreditHours: 9; hasName: Introduction to Quantum Physics; teaches: ProfHawkins
AIEthicsWorkshop	Workshop	hasCreditHours: 3; hasName: AI Ethics and Society Workshop

Research Projects, Grants & Resources

Individual Name	Class Type	Properties & Relationships
AuditoriumMagna	Auditorium	hasCapacity: 1200; hasName: Aula Magna, University of Verona
Project3DRecon	CollaborativeResearchProject	hasName: Mixed Reality Scene Reconstruction Using Zephyr and Unity; produces: PaperMixedReality
ComputerLabCV	ComputerLaboratory	hasCapacity: 30; hasName: Computer Vision Lab, Building A, University of Verona
PaperMixedReality	ConferencePaper	authors: ProfCastellani; hasName: Mixed Reality Interfaces Using 3D Reconstruction Pipelines
HorizonEuropeGrant	EUGrant	hasAmount: 250000.00; hasName: Horizon Europe Research Grant 2024
ErasmusMobilityGrant	ErasmusGrant	hasAmount: 4800.00; hasName: Erasmus+ Mobility Grant 2024-2025
ProjectClimateML	FundedResearchProject	fundedBy: HorizonEuropeGrant; hasName: Machine Learning for Climate Prediction
ProjectNeuralOperators	FundedResearchProject	conductsResearch: DrPatelPostdoc; hasName: Neural Operators for Partial Differential Equations
MIURGrant	GovernmentGrant	hasAmount: 80000.00; hasName: MIUR National Research Fund Grant
ProjectERM	InternalResearchProject	hasName: Entropic Risk Minimization for Portfolio Optimization
JournalDeepLearning	JournalArticle	authors: ProfGambara; hasName: Advances in Deep Learning for Financial Risk
LectureHallAlpha	LectureHall	hasCapacity: 200; hasName: Lecture Hall Alpha, Building A, University of Verona
LibraryMainVerona	Library	hasCapacity: 500; hasName: Biblioteca Centrale, University of Verona
MeritScholarshipAI	MeritScholarship	hasAmount: 3000.00; hasName: Merit Scholarship in Artificial Intelligence
NeedBasedBursaryVerona	NeedBasedScholarship	hasAmount: 2000.00; hasName: Need-Based Bursary, University of Verona
HCI_FinalExam	OralExam	hasGrade: 29.5; hasName: HCI Final Oral Exam, June 2025
PreprintRLTrading	Preprint	authors: DrPatelPostdoc; hasName: Reinforcement Learning Strategies for Algorithmic Trading
IndustryGrant_IBM	PrivateGrant	hasAmount: 40000.00; hasName: IBM Academic Research Grant

MLFinance_ProjectAssessment	ProjectAssessment	hasGrade: 30.0; hasName: ML in Finance Project 6 - Entropic Risk Minimization
LinearAlgebraQuiz	QuizAssessment	hasGrade: 9.5; hasName: Linear Algebra Weekly Quiz 3
ResearchFellowshipAILab	ResearchFellowship	hasAmount: 12000.00; hasName: AI Lab Research Fellowship
ResidenceHallEuropa	ResidenceHall	hasCapacity: 250; hasName: Residence Hall Europa, Verona
TechReportCV	TechnicalReport	authors: DrFerrari; hasName: Survey on Camera Calibration Methods for AR Systems
BiologyWetLab	WetLaboratory	hasCapacity: 20; hasName: Biology Wet Laboratory, Faculty of Medicine, Verona
PhysicsMidterm	WrittenExam	hasGrade: 88.0; hasName: Quantum Physics Midterm Written Exam

4. Ontology Design Analysis & Recommendations

The *Academic Organization Ontology* is a well-structured domain ontology conforming to OWL-DL standards. An analysis of its structural design reveals several highlights and suggestions for future extensibility:

- **Deep Taxonomy:** The taxonomy spans up to 4 inheritance levels (e.g., *AcademicConcept* → *Person* → *Student* → *GraduateStudent* → *MastersStudent*). This provides high granularity for categorizing objects.
- **Disjointness Axioms:** Crucial disjointness constraints are set up (e.g., *Student* is disjoint with *AcademicStaff* and *AdministrativeStaff*). This prevents logical contradictions (e.g., a person being both a student and staff at the same time unless roles are explicitly modeled).
- **Logical Domains/Ranges:** All 15 object properties and 8 datatype properties have defined `rdfs:domain` and `rdfs:range`. This strictly restricts properties to valid classes, facilitating consistency checks by semantic reasoners (e.g., HermiT or Pellet).
- **Rich Instance Set:** The dataset contains 81 naming individuals with highly interconnected cross-references (e.g., students enrolled in courses, supervised by professors, belonging to departments, and receiving scholarships).

Recommendations for Future Extensibility

- **Inverse Properties:** Define inverse properties for bilateral relationships (e.g., `hasMember` as inverse of `belongsTo`, and `hasStudent` as inverse of `enrolledIn`) to allow reasoners to infer double-sided links.
- **Transitive Properties:** Declare properties like `partOf` as transitive. If *CS Department* is `partOf` *Science Faculty*, and *Science Faculty* is `partOf` *University of Verona*, a reasoner can automatically infer that the department is `partOf` the university.
- **Role Modeling:** Instead of modeling permanent classes like *Student* or *Professor* directly as subclasses of *Person*, consider modeling them as 'Roles' a *Person* can hold over time. This supports scenario evolution, such as a student graduating and becoming a professor.